

- This exam contains 10 pages (including the front and back of this cover page) and 9 problems. Check to see if any pages are missing.
- This is the second exam and will count for 20% of your final grade. You have 105 minutes (5:15 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 220 points total in this exam, as well as 1 bonus questions worth a total of 5 possible points.

Page:	3	4	5	6	7	8	9	Total
Points:	20	30	30	30	40	30	40	220
Score:								

Bonus:	
--------	--

Review Day Question Bonus:	
----------------------------	--

1.

For readability in the key, correct answers are colored in red, e.g. **Correct Answer**, while incorrect answers are marked with a strike, e.g. ~~Incorrect Answer~~.

- (a) **2 points** A subspace of $\mathbb{R}^n$ is any set H contained in $\mathbb{R}^n$ such that:

(1) $0 \in H$, (2) H is closed under vector addition, (3) H is closed under scalar multiplication.

True~~False~~

- (b) **2 points** The columns of an invertible $n \times n$ matrix form a basis for $\mathbb{R}^n$.

True~~False~~

- (c) **2 points** The value of $\text{rank}(A)$ is the dimension of $\text{Nul}(A)$.

~~True~~**False**

- (d) **2 points** If $B = \{v_1, \dots, v_p\}$ is a basis for a subspace H , and if $x = c_1v_1 + \dots, c_pv_p$ for some $x \in H$, then the values of $c_1, \dots, c_p$ are the entries in the B -coordinate vector $[x]_B$.

True~~False~~

- (e) **2 points** If $B = \{v_1, \dots, v_p\}$ is a basis for a subspace H , then any vector x in H can be written in at least two different ways as a linear combination of the vectors in B .

~~True~~**False**

- (f) **2 points** If A is a triangular $n \times n$ matrix, then $\det(A)$ is the product of the entries on the main diagonal of A .

True~~False~~

- (g) **2 points** If the columns of A are linearly dependent, then $\det(A) = 0$.

True~~False~~

- (h) **2 points** The set of all solutions to a homogeneous linear system is the kernel of a linear transformation.

True~~False~~

- (i) **2 points** If B is an echelon form of a matrix A , then the pivot columns of B form a basis for $\text{Col}(A)$.

~~True~~**False**

- (j) **2 points** If A is an invertible matrix, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

True~~False~~

2. 30 points In this question, you are given a matrix A and an echelon form of A . Use this information to find a basis for $\text{Col}(A)$ and $\text{Nul}(A)$.

$$A = \begin{bmatrix} 4 & 5 & 9 & -2 \\ 6 & 5 & 1 & 12 \\ 3 & 4 & 8 & -3 \end{bmatrix} \xrightarrow{\text{Row Reduce}} \begin{bmatrix} 1 & 2 & 6 & -5 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Since the pivot columns of A form a basis for the column space of A (by Theorem 13 of section 2.8 in your textbook), the echelon form of A tells us that the first and second columns of A form

a basis for $\text{Col}(A)$. That is, the set $\left\{ \begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix} \right\}$ is a basis for $\text{Col}(A)$.

To find a basis for $\text{Nul}(A)$, we begin by solving the system $Ax = 0$. Since the solution set to this system is preserved under elementary row operations, we use the echelon form of A to produce the system:

$$\begin{aligned} x_1 + 2x_2 + 6x_3 - 5x_4 &= 0 \\ x_2 + 5x_3 - 6x_4 &= 0 \\ 0 &= 0. \end{aligned}$$

By the usual method of back substitution, we solve this system and find the parametric vector form of the solution set:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4x_3 - 7x_4 \\ -5x_3 + 6x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 4 \\ -5 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -7 \\ 6 \\ 0 \\ 1 \end{bmatrix}.$$

That is, the set $\left\{ \begin{bmatrix} 4 \\ -5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -7 \\ 6 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for $\text{Nul}(A)$.

3. 30 points Use determinants to decide if the set of vectors $\left\{ \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} -3 \\ -5 \\ -2 \end{bmatrix} \right\}$ is linearly independent.

Recall that the Invertible Matrix Theorem tells us that a set of n vectors $\{v_1, \dots, v_n\} \subseteq \mathbb{R}^n$ is linearly independent if and only if the matrix $A = [v_1 \ \dots \ v_n]$ (that is, the matrix with

$\{v_1, \dots, v_n\}$ as its columns) is invertible. So, we need only determine if the matrix $\begin{bmatrix} 4 & -7 & -3 \\ 6 & 0 & -5 \\ 2 & 7 & -2 \end{bmatrix}$

is invertible to determine if $\left\{ \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} -3 \\ -5 \\ -2 \end{bmatrix} \right\}$ is linearly independent. To that end,

recall that a square matrix A is invertible if and only if $\det(A) \neq 0$ (this is Theorem 4 in Section 3.2 of your textbook). Hence, we carry out the following computation:

$$\begin{aligned} \det \left(\begin{bmatrix} 4 & -7 & -3 \\ 6 & 0 & -5 \\ 2 & 7 & -2 \end{bmatrix} \right) &= (4)\det \left(\begin{bmatrix} 0 & -5 \\ 7 & -2 \end{bmatrix} \right) - (-7)\det \left(\begin{bmatrix} 6 & -5 \\ 2 & -2 \end{bmatrix} \right) + (-3)\det \left(\begin{bmatrix} 6 & 0 \\ 2 & 7 \end{bmatrix} \right) \\ &= 4(35) + 7(-2) - 3(42) \\ &= 0. \end{aligned}$$

Therefore, we may conclude $\left\{ \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} -3 \\ -5 \\ -2 \end{bmatrix} \right\}$ is not linearly independent.

4. 30 points Consider the polynomials $p_1(t) = 1 + t^2$ and $p_2(t) = 1 - t^2$. Is $\{p_1, p_2\}$ a linearly independent set in $\mathbb{P}_3$? Why or why not?

To determine whether $\{p_1, p_2\}$ is a linearly independent set in $\mathbb{P}_3$, it will be convenient to convert p_1 and p_2 to coordinate vectors in the standard basis $B = \{1, t, t^2, t^3\}$ of $\mathbb{P}_3$. Doing so, we yield

$$[p_1]_B = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad [p_2]_B = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}.$$

Since $[p_1]_B$ is not a scalar multiple of $[p_2]_B$, it follows that $\{[p_1]_B, [p_2]_B\}$ is a linearly independent set in $\mathbb{R}^4$. As a set of vectors $\{v_1, \dots, v_p\}$ belonging to a vector space V is linearly independent if and only if $\{[v_1]_B, \dots, [v_p]_B\}$ is linearly independent for any basis B of V , it follows that $\{p_1, p_2\}$ is a linearly independent set in $\mathbb{P}_3$.

5. 20 points The set $B = \left\{ \begin{bmatrix} 2 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$. Find the change-of-coordinates matrix from the basis B to the standard basis of $\mathbb{R}^2$.

Recall that the change of coordinates matrix from a basis B of $\mathbb{R}^n$ to the standard basis in $\mathbb{R}^n$ is the $n \times n$ matrix P_B such that $x = P_B[x]_B$. So, given a vector $x \in \mathbb{R}^2$, the B -coordinate vector of x is the vector $[x]_B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$ such that $x = c_1 b_1 + c_2 b_2$. Rearranging, this says $x = \begin{bmatrix} b_1 & b_2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$. Therefore, we may conclude that the change of coordinates matrix from the basis B to the standard basis of $\mathbb{R}^n$ is given by

$$P_B = \begin{bmatrix} b_1 & b_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -9 & 8 \end{bmatrix}.$$

6. 20 points Let $p_1(t) = 1 + t^2$, $p_2(t) = t - 3t^2$, and $p_3(t) = 1 + t - 3t^2$. The set $B = \{p_1, p_2, p_3\}$ forms a basis for $\mathbb{P}_2$. Find the polynomial $q \in \mathbb{P}_2$ given that $[q]_B = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$.

To find the polynomial $q \in \mathbb{P}_2$ such that $[q]_B = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$ for the basis $B = \{p_1, p_2, p_3\}$ of $\mathbb{P}_2$, we need only use the change-of-coordinates matrix $P_B = \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix}$ in the following manner:

$$\begin{aligned} q(t) &= P_B[q]_B \\ &= \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \\ &= -p_1(t) + p_2(t) + 2p_3(t) \\ &= -(1 + t^2) + (t - 3t^2) + 2(1 + t - 3t^2) \\ &= -1 - t^2 + t - 3t^2 + 2 + 2t - 6t^2 \\ &= 1 + 3t - 10t^2. \end{aligned}$$

7. 30 points You may assume that $B = \{b_1, b_2\}$ and $D = \{d_1, d_2\}$ are bases for $\mathbb{R}^2$. Find the change-of-coordinates matrix from B to D and the change-of-coordinates matrix from D to B .

$$b_1 = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \quad b_2 = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, \quad d_1 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}, \quad d_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

To find the change-of-coordinates matrix from B to D , we simply form the augmented matrix $\left[\begin{array}{cc|cc} d_1 & d_2 & b_1 & b_2 \end{array} \right]$ and row-reduce:

$$\left[\begin{array}{cc|cc} d_1 & d_2 & b_1 & b_2 \end{array} \right] = \left[\begin{array}{cc|cc} 1 & -2 & 7 & -3 \\ -5 & 2 & 5 & -1 \end{array} \right] \xrightarrow{\text{Row Reduce}} \left[\begin{array}{cc|cc} 1 & 0 & -3 & 1 \\ 0 & 1 & -5 & 2 \end{array} \right].$$

So, the change-of-coordinates matrix from B to D is given by $\begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}$.

To find the change-of-coordinates matrix from D to B , we could invert the change-of-coordinates matrix found above to produce

$$\begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 1 \\ -5 & 3 \end{bmatrix}.$$

Alternatively, we could simply form the augmented matrix $\left[\begin{array}{cc|cc} b_1 & b_2 & d_1 & d_2 \end{array} \right]$ and row-reduce:

$$\left[\begin{array}{cc|cc} b_1 & b_2 & d_1 & d_2 \end{array} \right] = \left[\begin{array}{cc|cc} 7 & -3 & 1 & -2 \\ 5 & -1 & -5 & 2 \end{array} \right] \xrightarrow{\text{Row Reduce}} \left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & -5 & 3 \end{array} \right].$$

So, using either approach, we may see that the change-of-coordinates matrix from D to B is given by

$$\begin{bmatrix} -2 & 1 \\ -5 & 3 \end{bmatrix}.$$

8. 40 points Suppose A is an $m \times n$ matrix with real entries and suppose $b \in \mathbb{R}^m$. What must be true of $\text{rank} \left(\begin{bmatrix} A & b \end{bmatrix} \right)$ and $\text{rank}(A)$ in order for the equation $Ax = b$ to be consistent?

Let $a_1, \dots, a_n \in \mathbb{R}^m$ be the columns of A so that $A = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix}$. Then, $Ax = b$ has a solution (that is, $Ax = b$ is consistent) if and only if there exists a vector $x \in \mathbb{R}^n$ such that

$$Ax = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + \dots + x_n a_n = b.$$

That is, the equation $Ax = b$ has a solution if and only if b is a linear combination of the columns of A . Said another way, if $Ax = b$ has a solution, then $b \in \text{Col}(A)$. But, since $\text{Col}(A) = \text{span}\{a_1, \dots, a_n\}$, then it must be the case that $\text{span}\{a_1, \dots, a_n, b\} = \text{span}\{a_1, \dots, a_n\}$ whenever $Ax = b$ has a solution. Therefore, noting that

$$\text{rank}(A) = \dim(\text{Col}(A)) = \dim(\text{span}\{a_1, \dots, a_n\})$$

and

$$\text{rank} \left(\begin{bmatrix} A & b \end{bmatrix} \right) = \dim(\text{Col} \left(\begin{bmatrix} A & b \end{bmatrix} \right)) = \dim(\text{span}\{a_1, \dots, a_n, b\}),$$

we may conclude that if $Ax = b$ is consistent, then

$$\text{rank}(A) = \text{rank} \left(\begin{bmatrix} A & b \end{bmatrix} \right).$$

9. 5 points (bonus) Determine whether or not A is invertible, where A is the 6×6 matrix

$$A = \begin{bmatrix} 1 & 2 & \pi^2 & 7 & \sqrt{2} & -9 \\ 0 & -1/2 & 0 & 1/19 & \sin(9) & 12 \\ 0 & 0 & 3 & 12/\pi & \tan(9) & -12 \\ 0 & 0 & 0 & 1/3 & (9) & 12 \\ 0 & 0 & 0 & 0 & -2 & -12 \\ 0 & 0 & 0 & 0 & 0 & 6 \end{bmatrix}.$$

Since the determinant of a triangular matrix is the product of its diagonal entries, we have

$$\det(A) = (1)(-1/2)(3)(1/3)(-2)(6) = 6.$$

So, as an $n \times n$ matrix M is invertible if and only if $\det(M) \neq 0$, we may conclude from the above calculation that A is invertible